New Wi-Fi system that can provide hundred times faster Internet

Scientists at Eindhoven University of Technology in The Netherlands have developed a new wireless Internet based on harmless infrared rays that is hundred times faster than existing Wi-Fi networks and has the capacity to support more devices without getting congested.

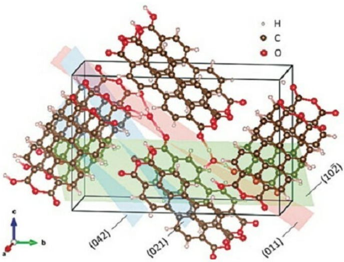
The new wireless network not only has a huge capacity—more than 40 Gigabits per second—there is no need to share since every device gets its own ray of light.

The system is simple and cheap to set up. Wireless data comes from a few central light antennae, for instance, mounted on the ceiling, which can very precisely direct the rays of light supplied by an optical fibre. The antennae contain a pair of gratings that radiate light rays of different wavelengths at different angles (passive diffraction gratings).

Changing the light wavelengths also changes the direction of the ray of light. Since a safe infrared wavelength is used that does not reach the vulnerable retina in your eye, this technique is harmless, too.

Researchers devise new hydronium-ion battery

A new type of battery developed by scientists at Oregon State University (OSU), USA, shows promise for sustainable, high-power energy storage. It is the world's first battery to use only hydronium ions as the charge carrier. The battery provides an additional option for researchers, particularly in the area of stationary storage.



New hydronium-ion battery presents opportunity for more sustainable energy storage (Image courtesy: http://oregonstate.edu)

Stationary storage refers to batteries in a permanent location that store grid power, including power generated from alternative energy sources such as wind turbines or solar cells, for use on a standby or emergency basis.

Hydronium, also known as H_3O^+, is a positively-charged ion produced when a proton is added to a water molecule. Researchers at OSU College of Science have demonstrated that hydronium ions can be reversibly stored in an electrode material consisting of perylenetetracarboxylic dianhydridem, or PTCDA. This material is an organic, crystalline, molecular solid. The battery, created in Department of Chemistry at OSU, uses dilute sulfuric acid as the electrolyte.

Lip-reading AI to aid those with hearing loss

Scientists at Oxford University in the UK, in collaboration with the company DeepMind, have developed a new artificial intelligence (AI) program that can lip-read more accurately than people—an advance that will help those who suffer from hearing loss.

The Watch, Attend and Spell (WAS) software system uses computer vision and machine learning methods to learn how to lip-read from a dataset made up of more than 5000 hours of TV footage, gathered from six different programmes.

The researchers compared the ability of the machine and a human expert to work out what was being said in the silent video. They found that the software was more accurate as compared to the professional. The human correctly read 12 per cent of words, while the software recognised 50 per cent of the words in the dataset, without error.

Brain-reading tech allows talking to paralysed victims

A brain-computer interface has been used in Switzerland to read thoughts and conduct basic communication with locked-in patients, who have absolutely no control over their body. The technology detects brain activity of the patients, who have not been able to speak for years, enabling them to answer yes-no questions. Accuracy of the system has been reported to be about 75 per cent.

Three women and one man, aged 24 to 76, were trained to use the system more than a year after they were diagnosed with completely locked-in syndrome, or CLIS. The condition was brought on by amyotrophic lateral sclerosis, or ALS, a progressive neurodegenerative disease that leaves people totally paralysed but still aware and able to think.